Application Messaging Services
Amazon SNS
Tutorial

Topics

1. Using Amazon Simple Notification Service (Amazon SNS) via the AWS SDK for Python

Prerequisite

- Complete this tutorial using the AWS Academy Learner Lab dedicated to our module[1].
- Note that if you are completing this tutorial using the AWS Academy Learner Lab, there is a restriction on the **AWS Regions** that can be used, namely all service access is limited to the **us-east-1** and **us-west-2** AWS Regions unless mentioned otherwise.
- This tutorial assumes that you are using the AWS Cloud9 Environment set up for developing in Python. Otherwise please complete the section *Setting up Cloud9 for Cloud Application Development in Python* of the Tutorial on AWS Cloud9 for Python.

Using Amazon Simple Notification Service via the AWS SDK for Python

Amazon Simple Notification Service (Amazon SNS)[2] is a fully managed messaging service that provides message delivery from publishers to subscribers. "Publishers communicate asynchronously with subscribers by sending messages to a topic, which is a logical access point and communication channel. Clients can subscribe to the SNS topic and receive published messages using a supported protocol, such as Amazon SQS, AWS Lambda, HTTP, email, mobile push notifications, and mobile text messages (SMS)." (source: https://docs.aws.amazon.com/sns/latest/dg/welcome.html).

[1] This assumes that you have accepted the invitation to join the AWS Academy Learner Lab received via your NCI email account.

[2] https://aws.amazon.com/sns

Objectives

The aim of this tutorial is to learn how to programmatically use Amazon SNS. More specifically, at the end of this tutorial you should know to perform the following tasks:

- Create a topic
- Publish a message to an Amazon SNS topic
- Subscriptions
  - Subscribe to an Amazon SNS topic
  - Unsubscribe from an Amazon SNS topic
- Delete an Amazon SNS topic

Recall that the AWS SDKs provide two levels of APIs for the AWS services, namely Client (or "low-level") APIs and Service Resource APIs. Please consult the documentation of Amazon SNS available at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/sns.html.

In this tutorial we will use the SNS Client which is a low-level client representing the Amazon Simple Notification Service (Amazon SNS). Please consult the documentation of Amazon SNS Client available at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/sns.html#client.

First, consult the documentation of Amazon SNS Client to identify the available methods which enable us to perform the tasks listed under the *Objectives* section.

Sample Python Program The folder *messaging_sns_demos* available on the module's Moodle page shows examples of how to perform core operations with Amazon SNS programmatically via the available AWS SDK for Python.

Running the sample Python program

Note that before we can run the demo, we first need to create a topic, and, then, in this example, a subscriber (a queue in this demo) would subscribe to the created topic. To see the available arguments please use

```
python sns_demo_setup.py  --help
```

For example, if we want to create a topic named "mynews", and to create the queue named "mydemoqueue" which subscribes to the topic, we would run the following comand at the terminal

```
python sns_demo_setup.py mynews --create_topic --setup_queue --queue_name mydemoq
```

In order to run the demo, first, check the available arguments by running the following command at the terminal

```
python sns_demo.py --help
```

Task: Consult the Python Code Samples for Amazon SNS available at https://docs.aws.amazon.com/code-library/latest/ug/python_3_sns_code_examples.html.

Resources

- Amazon SNS https://aws.amazon.com/sns
- Amazon Simple Notification Service Developer Guide https://docs.aws.amazon.com/sns/latest/dg/welcome.html
- Amazon Simple Notification Service API Reference https://docs.aws.amazon.com/sns/latest/api/Welcome.html
- Boto3 Amazon SNS Documentation https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/sns.html
- Amazon SNS examples using SDK for Python (Boto3) https://docs.aws.amazon.com/code-library/latest/ug/python_3_sns_code_examples.html
- AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/welcome.html
- Python Tutorial for AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/sample-python.html